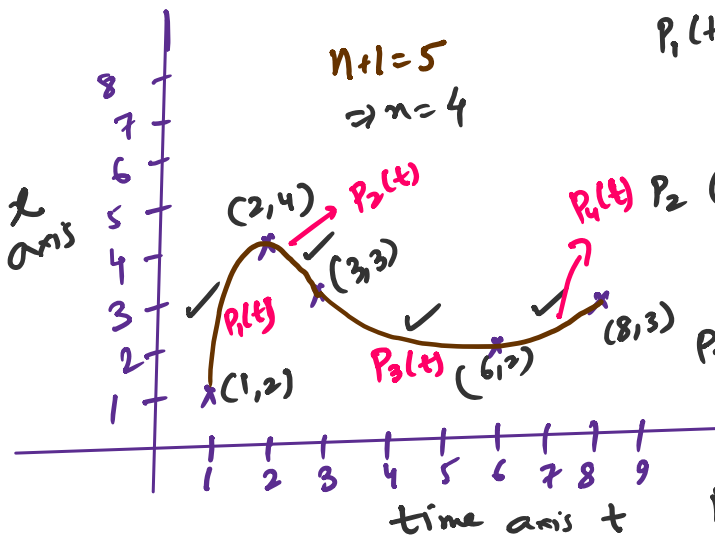


Cubic Spline Interpolation Example

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11:19 AM



$$P_1(t) = a_1(t-t_1)^3 + b_1(t-t_1)^2 + c_1(t-t_1) + d_1$$

$$P_2(t) = a_2(t-t_2)^3 + b_2(t-t_2)^2 + c_2(t-t_2) + d_2$$

$$P_3(t) = a_3(t-t_3)^3 + b_3(t-t_3)^2 + c_3(t-t_3) + d_3$$

$$P_4(t) = a_4(t-t_4)^3 + b_4(t-t_4)^2 + c_4(t-t_4) + d_4$$

$i = 1 \text{ to } n$

First Condition.

$$\begin{aligned} d_1 &= x_1 = 2 \\ d_2 &= x_2 = 4 \\ d_3 &= x_3 = 3 \\ d_4 &= x_4 = 2 \end{aligned}$$

$$h_i = t_{i+1} - t_i$$

$$h_1 = t_2 - t_1 = 2 - 1 = 1$$

$$h_2 = t_3 - t_2 = 3 - 2 = 1$$

$$h_3 = t_4 - t_3 = 6 - 3 = 3$$

$$h_4 = t_5 - t_4 = 8 - 6 = 2$$

Use the second Condition.

$$P_2(t_2) = P_1(t_2)$$

$$\Rightarrow a_2(t_2 - t_2)^3 + b_2(t_2 - t_2)^2 + c_2(t_2 - t_2) + d_2$$

$$= a_1(t_2 - t_1)^3 + b_1(t_2 - t_1)^2 + c_1(t_2 - t_1) + d_1$$

$$\Rightarrow d_2 = a_1 + b_1 + c_1 + 2 \Rightarrow a_1 + b_1 + c_1 = 2$$

From Clamped Boundary Condition.

$$\textcircled{1} d_i = x_i \quad i = 1 \text{ to } n \quad d_1 = 2 \quad d_2 = 4 \quad d_3 = 3 \quad d_4 = 2$$

$$h_1 = 1 \quad h_2 = 1 \quad h_3 = 3 \quad h_4 = 2$$

$$\textcircled{2} (2b_1 + b_2)h_1 = \frac{3(d_2 - d_1)}{h_1} - 3p$$

$$\textcircled{3} b_{i-1}h_{i-1} + 2b_i(h_i + h_{i-1}) + b_{i+1}h_{i+1}$$

$$\textcircled{3} \quad b_{i-1} h_{i-1} + 2b_i (h_i + h_{i-1}) + b_{i+1} h_i$$

$$= 3 \left(\frac{d_{i+1} - d_i}{h_i} - \frac{d_i - d_{i-1}}{h_{i-1}} \right) \quad \downarrow \quad i = 2 \text{ to } n$$

$$\textcircled{4} \quad (2b_{n+1} + b_n)h_n = - \frac{(d_{n+1} - d_n)}{h_n} + 3q$$

$$\textcircled{5} \quad c_i = \frac{d_{i+1} - d_i}{h_i} - \frac{1}{3} (2b_i + b_{i+1})h_i$$

$$\textcircled{6} \quad a_i = \frac{b_{i+1} - b_i}{3h_i} \quad \text{Assume } p=0, q=0$$

$$(2b_1 + b_2) \cdot 1 = 3 \frac{(4-2)}{1} - 3 \times 0 = 6$$

$$\Rightarrow \boxed{2b_1 + b_2 = 6}$$

$$b_1 h_1 + 2b_2 (h_2 + h_1) + b_3 h_2 \quad i=2$$

$$= 3 \left(\frac{d_3 - d_2}{h_2} - \frac{d_2 - d_1}{h_1} \right)$$

$$\Rightarrow b_1 + 4b_2 + b_3 = 3 \left(\frac{3-4}{1} - \frac{4-2}{1} \right)$$

$$\Rightarrow b_1 + 4b_2 + b_3 = 3(-1-2)$$

$$\Rightarrow \boxed{b_1 + 4b_2 + b_3 = -9}$$

$i=3$

$$b_2 h_2 + 2b_3 (h_3 + h_2) + b_4 h_3$$

$$= 3 \left(\frac{d_4 - d_3}{h_3} - \frac{d_3 - d_2}{h_2} \right)$$

$$b_2 + 8b_3 + 3b_4 = 3 \left(\frac{2-3}{3} - \frac{3-4}{1} \right)$$

$$b_2 + 8b_3 + 3b_4 = 3 \left(\frac{2-3}{3} - \frac{3-4}{1} \right)$$

$$b_2 + 8b_3 + 3b_4 = 3 \left(-\frac{1}{3} + 1 \right) = -1 + 3 = 2$$

$$b_2 + 8b_3 + 3b_4 = 2$$

$$i=4$$

$$b_3 h_3 = 2b_4 (h_4 + h_3) + b_5 h_4 = 3 \left[\frac{d_5 - d_4}{h_4} - \frac{d_5 d_3}{h_3} \right]$$

$$d_5 = 3$$

$$3b_3 + 2b_4 (2+3) + 2b_5 = 3 \left(\frac{3-2}{2} - \frac{2-3}{3} \right)$$

$$\Rightarrow 3b_3 + 10b_4 + 2b_5 = 3 \left(\frac{1}{2} + \frac{1}{3} \right) = \frac{5}{2}$$

$$(2b_5 + b_4) h_4 = - \frac{(d_5 - d_4)}{h_4} + 3 \cdot 0$$

$$\rightarrow (2b_5 + b_4) \cdot 2 = -3 \left(\frac{3-2}{2} \right) = -\frac{3}{2}$$

$$\Rightarrow 4b_5 + 2b_4 = -\frac{3}{2}$$

$$\underbrace{\begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 1 & 4 & 1 & 0 & 0 \\ 0 & 1 & 8 & 3 & 0 \\ 0 & 0 & 3 & 10 & 2 \\ 0 & 0 & 0 & 2 & 4 \end{bmatrix}}_A \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \end{bmatrix}}_B = \underbrace{\begin{bmatrix} 6 \\ -9 \\ 2 \\ 5/2 \\ -3/2 \end{bmatrix}}_Y$$

$$AB = Y \Rightarrow A^{-1}AB = A^{-1}Y$$

$$IB = A^{-1}Y \Rightarrow B = A^{-1}Y$$

$$B = A/Y$$

$$(i+1)^{100} \Rightarrow i+1 = \sqrt{2} \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right)$$

$$\left[\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) \right]^{100}$$

$$2^{50} \left[\cos 25\pi + i \sin 25\pi \right]$$

$$\Rightarrow 2^{50} \times -1 = -1 \cdot 2^{50}$$

